

Phase 8: Business Case Document

Section 1: Problem

This document presents the business case for Pharma-Pulse: a predictive intelligence system that provides 8–12 weeks of early warning on generic drug shortages in India's public health procurement segment.

The business case answers three questions that any decision-maker will ask before committing to a solution:

- What is this problem actually costing, in money and in lives?
- What does it cost to solve it?
- Is solving it worth it?

The audience for this document is a state health agency procurement head, a district health officer, or any institutional decision-maker evaluating whether to pilot or adopt Pharma-Pulse.

2.1 Scale of Public Health Procurement in India

India's public health system manages procurement of essential medicines across 28 states and 8 union territories. The scale is significant:

- The Government of India spends approximately **₹8,000–10,000 crore annually** on medicines under the National Health Mission
- A single state like Maharashtra or Tamil Nadu manages annual medicine procurement budgets in the range of **₹400–800 crore**
- Each district procurement officer manages a portfolio of **300–500 essential medicines** across multiple health facilities

Source: National Health Mission financial reports, GeM public procurement data

2.2 The Cost of a Single Shortage Event

This is the number most people have never calculated. Let us build it from the ground up.

A shortage event is defined as a period where a drug is unavailable at point of care for 7 days or more at a government facility.

Direct procurement cost:

When a shortage occurs, the government must either conduct emergency procurement, paying significantly above the tender rate, or allow the drug to remain unavailable. Emergency procurement outside normal tender cycles typically costs **30–60% above the standard contracted rate**.

For a drug like Amoxicillin 500mg, the standard GeM rate is approximately ₹4–6 per tablet. Emergency procurement outside tender cycles pushes this to ₹6–9 per tablet. For a district hospital consuming 50,000 tablets per quarter, that price differential alone represents a **₹1.5–3 lakh additional cost per event**.

Staff time cost:

The IJMR paper (Vashisht et al., 2025) found that procurement staff spend significant time managing shortage responses, manual supplier calls, re-tendering, substitute identification, documentation. A

conservative estimate of 40 staff-hours per shortage event at a district level, valued at a government officer rate, represents approximately **₹15,000–25,000 per event in absorbed labour cost.**

Clinical substitution cost:

When a drug is unavailable, clinicians prescribe therapeutic substitutes. These substitutes are frequently more expensive. WHO (2016) documented that US hospitals incurred **\$416 million annually**, \$200 million on expensive alternatives and \$216 million in labour costs, from shortage management. Applying a conservative India-adjusted ratio, a single state-level shortage event for a widely-used generic generates an estimated **₹5–15 lakh in substitution and management costs.**

Patient outcome cost:

This is the hardest to quantify and the most important. When a medication gap occurs:

- Treatment delays extend hospital stays, average **2–4 additional bed-days** per affected patient at government hospital costs of ₹800–1,200 per bed-day
- Therapeutic substitution causes adverse drug events in approximately **3–5% of substitution cases**
- In critical care settings, medication gaps cause measurable deterioration in patient outcomes

For a district hospital serving 500 inpatients per day with a 7-day shortage of a critical antibiotic, even a conservative adverse outcome rate produces **50–80 patients receiving suboptimal care.**

2.3 Frequency of Shortage Events

The IJMR paper (2025) found a **53.9% shortfall** in medicine procurement through GeM at a studied government hospital over 12 months, meaning more than half of demanded medicines did not arrive in full. This is not a rare event. It is the baseline operating condition of public health procurement in India.

Based on available evidence:

- A typical district-level government hospital experiences **8–15 significant shortage events per year** across its essential medicines portfolio
 - A state-level procurement agency managing multiple districts experiences the **aggregate impact of hundreds of facility-level shortage events annually**
 - The 2026 FOPE warning (Bajeli-Datt, The New Indian Express, March 2026) documented an **imminent national-scale shortage** of four essential medicines simultaneously, driven by upstream signals that were detectable weeks earlier
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2.4 Total Annual Cost of Inaction, Per State

Cost Component	Per Event Estimate	Events Per Year	Annual Cost Estimate
Emergency procurement premium	₹2-4 lakh	12	₹24-48 lakh
Staff time and administrative cost	₹0.25 lakh	12	₹3-4 lakh
Substitution and clinical management	₹8-12 lakh	12	₹96-144 lakh
Extended bed-days and adverse outcomes	₹5-10 lakh	12	₹60-120 lakh
Total annual cost of shortage			₹1.8-3.2 crore per state

Section 3: The Cost of the Solution

Pharma-Pulse is a proof-of-concept system built using entirely free public data sources and open-source tools. Its cost structure is unusually lean.

3.1 Development Cost (One-Time)

Activity	Estimated Cost
BA lifecycle documentation (9 phases)	Staff time — analyst salary equivalent ₹3-4 lakh for 6 months
Python model development	Included in analyst time
Dashboard build (Power BI/Tableau)	Free tools — no licence cost
Data collection and structuring	Staff time — included above
Total one-time development cost	₹3-4 lakh (analyst time only)

3.2 Operating Cost (Annual)

Activity	Estimated Annual Cost
Weekly data refresh and model run	2 hours/week analyst time — ₹60,000-80,000 per year
Dashboard maintenance and updates	4 hours/month — included above
Data source monitoring (new sources, format changes)	8 hours/month — included above
Power BI or Tableau licence (if institutional)	₹0 (Tableau Public free) to ₹80,000 (Power BI Pro institutional)
Total annual operating cost	₹60,000-1,60,000 per year

3.3 Cost to Scale

Pharma-Pulse Version 1.0 covers 6 drugs and 12 states. Scaling to the full essential medicines list (400+ drugs) and all 36 states would require:

- Expanded data collection , primarily more CDSCO and GeM data extraction
- Additional analyst time for model retraining
- Institutional server or cloud hosting for automated pipeline runs

Estimated scaling cost to full national deployment: ₹8–15 lakh one-time, ₹3–5 lakh annual operating.

This is still a fraction of what a conventional government IT procurement would cost for an equivalent system.

Section 4: Return on Investment

4.1 The Core Calculation

The business case rests on one question: if Pharma-Pulse prevents even a fraction of the shortage events that currently occur, does the saving exceed the cost of running it?

Let us work through it conservatively.

Assumption: Pharma-Pulse prevents or significantly mitigates 3 shortage events per year at the state level by enabling pre-emptive procurement action on drugs flagged at high or critical risk with sufficient lead time.

This is a deliberately conservative assumption. The model generates 8–12 weeks of lead time. A procurement officer who acts on a high-risk flag has enough time to initiate a tender, pre-qualify an alternate supplier, or build buffer stock. Preventing 3 events out of 12 is a 25% reduction , a modest target.

	Conservative	Moderate	Optimistic
Shortage events prevented per year	3	5	8
Average cost per prevented event	₹15 lakh	₹18 lakh	₹20 lakh
Annual saving	₹45 lakh	₹90 lakh	₹1.6 crore
Annual operating cost of PharmaPulse	₹1.6 lakh	₹1.6 lakh	₹1.6 lakh
Net annual benefit	₹43.4 lakh	₹88.4 lakh	₹1.58 crore
ROI	27x	55x	99x

Even under the most conservative assumption , preventing just 3 shortage events per year , Pharma-Pulse returns **27 rupees for every 1 rupee invested in operating it.**

4.2 Break-Even Analysis

The system reaches break-even , meaning it has saved more than it cost to build and operate , the moment it prevents **its first significant shortage event.**

One prevented shortage event saving ₹15 lakh recovers the entire annual operating cost of ₹1.6 lakh with ₹13.4 lakh to spare.

The development cost of ₹3–4 lakh is recovered within the first quarter of operation if one medium-scale shortage event is prevented.

4.3 Beyond Money , The Outcomes You Cannot Put A Number On

The financial ROI is compelling. But it understates the true value of the system because it cannot fully account for:

Patient outcomes. A post-surgery patient in Nagpur who receives their antibiotic on time because the procurement officer was warned 9 weeks earlier does not appear in any cost calculation. The adverse event that did not happen generates no data. This invisibility is precisely why shortage prevention has been chronically underinvested in.

Procurement officer effectiveness. A district procurement officer managing 400 medicines reactively , responding to empty shelves , operates in a permanent state of crisis. The same officer, equipped with a Monday morning risk dashboard, can spend their time on strategic procurement decisions rather than emergency firefighting. That shift in operating mode has compounding benefits that no single ROI calculation captures.

Institutional credibility. State health agencies that demonstrate forward-looking procurement intelligence , preventing shortages before they become public , build institutional trust that has long-term value beyond any individual drug shortage event.

Section 5: Risk Assessment

5.1 Risks of Adopting Pharma-Pulse

Risk	Likelihood	Impact	Mitigation
Model generates false positives — flags drugs that do not actually run short	Medium	Low — wasted procurement effort on buffer stock that turns out unnecessary	Alert threshold calibration. False positives cost buffer stock spend, not shortage outcomes. Conservative thresholds reduce this.
Data source becomes unavailable — GeM or CDSCO changes portal structure	Low-Medium	Medium — one feature loses signal until source is updated	Six-feature model means no single source failure collapses the system. Model degrades gracefully.
Procurement officer does not act on alerts	Medium	High — system is only valuable if acted upon	Dashboard design built for non-technical users. Recommendations are actionable, not just informational. Change management required at institutional level.
Model accuracy insufficient for critical decisions	Low	High	Backtest validation against 4 known shortage events documented. Precision score above 0.80. Model presents confidence level with every score — not false certainty.

5.2 Risk of NOT Adopting PharmaPulse

Risk of Inaction	Current Likelihood	Impact
Continuation of 53.9% medicine procurement shortfall	High — this is the documented baseline	₹1.8-3.2 crore annual cost per state, ongoing
Next FOPE-scale national shortage event	Medium — geopolitical instability persists	Multi-state shortfall of 4+ essential medicines simultaneously
Patient harm from preventable medication gaps	High — occurring continuously at ward level	Clinical adverse outcomes, treatment delays, patient out-of-pocket costs
No progress toward WHO 2016 shortage notification system recommendation	Certain without intervention	India remains without national shortage intelligence infrastructure

Section 6: Recommendation

6.1 Immediate Recommendation

Pilot Pharma-Pulse at one state procurement agency for one quarter.

The pilot requires:

- Access to one state's GeM procurement data at category level
- A single procurement officer as the primary system user
- Weekly model refresh run by the analyst
- Post-quarter review comparing flagged drugs against actual shortage events

Cost of pilot: negligible. It runs on existing infrastructure using free tools and public data.

Expected pilot outcome: at least 2–3 high-risk flags that can be cross-validated against procurement records, demonstrating the system's predictive accuracy in a live environment.

6.2 Scale Recommendation

Following a successful pilot, Phase 2 deployment covers:

- Expansion to full state essential medicines list
- Integration with the state's existing ERP or HIS for automated stock data input
- Formal training for district procurement officers on dashboard interpretation
- Quarterly model retraining as new data accumulates

Pharma-Pulse costs ₹1.6 lakh per year to operate and returns ₹45 lakh in prevented shortage costs under the most conservative assumptions, a 27x return, while addressing a gap that the WHO identified as a global health system priority in 2016 and that no system currently fills for India's public health procurement segment.

Section 7: Sources Used in This Business Case

Claim	Source
53.9% medicine procurement shortfall	Vashisht et al. (2025). Indian J Med Res, 161, 32–37
US hospital shortage cost \$416 million annually	WHO Drug Information Vol. 30, No. 2 (2016)
FOPE national shortage warning, 4 essential drugs	Bajeli-Datt, K. New Indian Express, March 17, 2026
GeM procurement cycle 5–6 months	Vashisht et al. (2025); CMHO Phalodi Tender Document (2025)
WHO shortage notification system recommendation	WHO Drug Information Vol. 30, No. 2 (2016)
NHM medicine procurement expenditure	National Health Mission Annual Report, Government of India